Efficient Sub-Character Representation Learning and Transfer for Chinese Name Gender Prediction

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Abstract

Chinese character representations often treat logographic characters as atomic vocabulary tokens, ignoring the rich internal configurations embedded within sub-character components. While radicals provide localized semantic categories, Ideographic Description Sequences (IDS) preserve the explicit graphic positioning and spatial assembly order of those units. This paper introduces a highly efficient, lightweight transfer learning framework built entirely around ordered sub-character orthography. In the self-supervised phase, we train a non-recurrent 1D Convolutional Autoencoder paired with a 1D Transposed Convolutional Decoder to compress both radical identity and ordered structural decomposition strings into a compact, reusable 64-dimensional bottleneck vector. This representation is learned exclusively through structural reconstruction without using downstream task labels. The trained encoder is subsequently frozen and transferred to a name-gender prediction task over a large-scale dataverse CnGender dataset containing approximately one million records. Rather than deploying traditional vector average pooling which dilutes semantic cues, we implement a position-preserving feature concatenation layer to retain sequence-order alignment. In a preliminary evaluation on 20,000 records from the CnGender dataset, the frozen representation achieved 92.17% accuracy and an ROC-AUC of 0.9683. Operated via a lazy-evaluation data generator stream, the system runs efficiently on consumer-grade hardware.

Keywords: Chinese Character Representation, Sub-Character Features, Ideographic Description Sequence, 1D Convolutional Autoencoder, Transfer Learning, Tensorflow, Keras 3

1. Introduction

Unlike alphabetic languages that arrange phonemic characters linearly, Chinese orthography is logographic, organizing graphical units within a strict two-dimensional bounding grid. Each Hanzi character is constructed from recurring sub-character components arranged according to a small set of structural relationships. Unicode's Ideographic Description Sequence (IDS) [7] standard formalizes this topology by combining operand characters with Ideographic Description Characters (IDCs)—such as ⿰ for left-to-right division and ⿱ for top-to-bottom layout stacked configurations.

Conventional Chinese Natural Language Processing models map complete characters directly to unique vocabulary tokens. While effective for high-frequency words, this design creates acute data sparsity when encountering rare, historical, or newly emerged characters, presenting a strict Out-of-Vocabulary (OOV) wall. Furthermore, token-level networks are completely blind to internal morphological components; they fail to recognize that an unindexed character has semantic ties to fluids due to its structural water radical (氵).

To bypass this, researchers have turned to sub-character configurations. Prior studies have deployed radical-enhanced or component-enhanced text vectors [1,2], while others have integrated raw character structural matrices directly into downstream language layers [3,4]. However, these approaches routinely rely on Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) layers to read these short sequence strings. LSTMs introduce a major sequential processing bottleneck, slow down backpropagation, and demand massive memory infrastructure when dealing with millions of lines of text.

The present work investigates a narrower, practical question: Can ordered Chinese structural information be compressed into a tiny, reusable vector learned once through self-supervision and transferred to an independent downstream task without encoder fine-tuning?

Our contributions are formalized as follows:

1. We design a parallelized 1D Convolutional Autoencoder (1D-CAE) that compresses a character's radical and ordered structural strings into a compact 64-dimensional latent bottleneck without sequential recurrent overhead.
2. We demonstrate a Position-Preserving Feature Concatenation paradigm for multi-character tasks, proving that retaining coordinate alignment completely eliminates the semantic feature dilution common in vector pooling.
3. We optimize a lazy-evaluation data generator stream that permits continuous, high-performance training over approximately one million naming records on local, memory-constrained hardware with minimum RAM degradation.

2. Methodology

The framework implements a two-stage learning paradigm separating representation compression from downstream prediction task learning.

Stage 1: Self-Supervised Structural Representation Learning

```
Radical + Ordered IDS Sequence --> [1D Conv Encoder] --> Bottleneck (64-D) --> Dual Reconstruction Loss
```

Stage 2: Position-Preserving Frozen Transfer Learning

```
Character 1 --> [Frozen Pretrained Encoder] --> 64-D Vector -+
                                                             |--> [Concatenate] --> 128-D Name Vector --> Sigmoid Pred
Character 2 --> [Frozen Pretrained Encoder] --> 64-D Vector -+
```

For the purpose of practical application, the second stage is used to infer gender from Chinese names.

Architecture Diagram

```
Radical + Ordered IDS
        |
        v
 1D Conv Encoder
        |
        v
      64-D
        |
        v
reconstruction (autoencoder)
        |
        | frozen
        v
Character 1 --> 64-D --+
                       |--> 128-D --> gender classifier
Character 2 --> 64-D --+
```

2.1 Stage 1: Self-Supervised Structural Autoencoder

The target character is broken down into two parallel visual features: its primary radical and its ordered structural decomposition sequence. We frame the structural layout string as a short, fixed-length sequence ($T_{max} = 12$) processed character-by-character natively via a custom Unicode splitting function within the model graph. Crucially, rather than processing these inputs in isolation or as a single combined raw text string, the framework passes the radical and the structural sequence through two independent embedding spaces. The categorical radical identity is mapped to a continuous dense vector, which is then concatenated along the feature dimension with the temporal embedding matrix generated by the sequence vectorizer. This fusion layer constructs a unified, dense representation matrix $X_{seq} \in \mathbb{R}^{T \times D_{in}}$ that simultaneously encapsulates localized semantic category markers and explicit sequence-order topology before entering the convolutional encoder. This ensures a sequence like □ | ? is parsed cleanly as the explicit array [□, | , ?] rather than an unparsed whitespace string.

Let $X_{seq} \in \mathbb{R}^{T \times D_{in}}$ represent the sequence matrix after embedding lookup. The Encoder Layer applies a parallel temporal 1D Convolution:

$$H = \text{ReLU}(\text{Conv1D}(X_{seq}, W_e) + b_e)$$

To collapse the temporal sequence dimensions without losing localized features, we pass the matrix through a non-parametric Global Average Pooling layer, followed by a dense projection to the bottleneck space:

$$Z_{latent} = \text{Dense}(\text{GlobalAveragePooling1D}(H)) \in \mathbb{R}^{64}$$

The Decoder Layer splits into a dual-branch output to verify structural reconstruction validity. The Radical Reconstruction Branch projects the latent vector through a Dense layer to predict the discrete categorical index of the radical:

$$\hat{r}_c = \text{Softmax}(\text{Dense}(Z_{latent}))$$

The Structural Sequence Reconstruction Branch projects the identical latent vector back to sequence dimensions and applies a 1D Transposed Convolution to compute categorical cross-entropy probabilities across the entire token vocabulary for each step:

$$H_{dec\_reshape} = \text{Reshape}(\text{Dense}(Z_{latent})) \in \mathbb{R}^{T \times D_{channels}}$$

$$\hat{X}_{seq} = \text{Softmax}(\text{Conv1DTranspose}(H_{dec\_reshape}, W_d) + b_d)$$

2.2 Multi-Target Loss Objective

The entire autoencoder network optimizes a joint multi-target objective function combining both categorical reconstruction branches:

$$\mathcal{L}_{total} = \mathcal{L}_{radical} + 2.0 \cdot \mathcal{L}_{sequence}$$

Both branches utilize sparse categorical cross-entropy, with the structural sequence pathway receiving double the loss weight to heavily prioritize the preservation of ordered layout topology.

2.3 Stage 2: Position-Preserving Concatenation Classifier

To evaluate the downstream utility of our frozen representations, we apply the encoder to the specific task of predicting gender based on a raw Chinese name string.

When extracting multi-character given name semantics, standard architectures typically pass individual character embeddings into an average pooling or summing block to arrive at a global phrase representation. We hypothesize that vector pooling can introduce semantic signal dilution by collapsing character-specific features into a single averaged representation. In Chinese given names, a highly gender-neutral character (e.g., “云”) combined with a strong feminine marker (e.g., “霞”) will have its gender probability washed out under average pooling layers.

To solve this, we preserve the encoder weights learned during Stage 1, freeze them completely (trainable = False), and extract character representations independently:

$$Z_1 = E_{frozen}(c_1) \in \mathbb{R}^{64}, \quad Z_2 = E_{frozen}(c_2) \in \mathbb{R}^{64}$$

The global name-level representation is constructed by concatenating the vectors in their original character sequence order:

$$Z_{Name} = \text{Concatenate}(Z_1, Z_2) \in \mathbb{R}^{128}$$

By mapping features into separate coordinate positions, downstream dense classification layers can explicitly learn that feminine markers occurring at index position 2 hold distinct semantic weight compared to index position 1. The final dense layers project features through a Sigmoid activation to predict the continuous target probability ratio $Y \in [0.0, 1.0]$.

3. Large-Scale Data Pipeline Optimization

Processing a 1-million-row dataset (CnGender.txt from the large-scale, peer-reviewed Chinese Name-to-Gender Dataset [6]) typically strains system RAM due to the overhead of managing large nested Python text objects in memory. We eliminate this performance barrier by constructing an asynchronous streaming dataset pipeline using `tf.data.Dataset.from_generator`.

The generator maintains a single open text pointer to the disk storage file, parsing incoming rows sequentially using a Tab () delimiter schema. Extracted characters are cross-referenced against a precompiled in-memory dictionary cache containing our pre-vectorized integer tensor maps. Missing characters are seamlessly routed to standardized fallback and structural padding blocks. By configuring the thread executor to `prefetch(tf.data.AUTOTUNE)`, the system runs with perfect background parallel processing: the CPU threads parse and vector-map batch $N + 1$ from disk while the GPU simultaneously executes gradient backpropagation on batch N.

To ensure complete empirical transparency and support technical reproducibility, our complete Tensorflow/Keras 3 source architectures, tokenization layers, and saved embedding matrices are hosted openly in our replication library [8].

4. Experiments and Results

4.1 Hyperparameters and Experimental Setup

The structural orthographic inventory was compiled using a standard, open-source Hanzi decomposition dataset [5] containing approximately 9,574 core standard characters to calibrate the base spatial encoder matrix. The network hyperparameters were standardized to a maximum sequence length of 12, a 32-dimensional radical embedding, and a 64-dimensional structural token embedding. The core encoder projects to a 64-dimensional latent bottleneck using a 64-filter Conv1D layer with a kernel size of 3. The model was optimized using Adam with a batch size of 512 over 10 epochs on an Apple Silicon environment (M-series processing architecture).

4.2 Performance Metrics

Because the target labels from the dataverse source are continuous gender probability ratios, we evaluate true binary classification performance by thresholding the ground truth at 0.5. The current evaluation uses a 20,000-record evaluation slice from the CnGender dataset. Because the current pipeline does not establish a fully independent held-out test protocol, these results should be regarded as preliminary.

Crucially, the evaluation dataset features a heavy class imbalance, containing 15,044 Male entries (~75.22%) and 4,956 Female entries (~24.78%). Because raw accuracy can be artificially inflated in skewed sets by predicting the majority class, we evaluate both Classification Accuracy and the Area Under the Receiver Operating Characteristic curve (AUC-ROC) to confirm true discriminative stability.

System Performance Metrics Dashboard

Evaluation Metric	Achieved Value	Performance Context
Classification Accuracy	92.17%	Global Naming Allocation Rate
ROC Area Under Curve (AUC)	0.9683	True Discriminative Stability under Class Imbalance

The model achieves a Classification Accuracy of 92.17% and an AUC-ROC of 0.9683. The ROC-AUC provides evidence that the frozen sub-character embedding representation retains substantial discriminative information for the downstream task despite the class imbalance, e.g., the minority-class gender-associated structural features rather than biasing toward majority-class distributions. Whether this discriminative information specifically arises from structural markers can be tested through the proposed ablation experiments.

4.3 Qualitative Error Analysis

We hypothesize that the discrepancies between the ground-truth ratios and model predictions primarily cluster around three socio-linguistic phenomena:

1. Historical Semantics vs. Modern Orthography (Etymological Shift): The autoencoder parses pure graphic morphology. For example, the radical 月 (moon) frequently represents 肉 (flesh/body parts) historically due to structural character merging over centuries (e.g., 肌, 肝, 胜). The network occasionally misinterprets these as cosmic or neutral tokens, skewing masculine names toward the class mean.
2. Phonetic-Semantic Clashes: Because our streamlined system operates purely on logographic geometry without explicit phonetic inputs (Pinyin features), characters used strictly for auditory resonance create noise. In a name like "英豪", if "英" is selected purely for its sound rather than its semantic connection to 艹 (vegetation, which skews feminine), the encoder overweights the floral radical and lowers male probability despite the presence of the highly masculine "豪" component.
3. Sociological Inversions (Gender-Reversed Stylistic Names): Cultural naming behaviors occasionally contradict standard orthographic skews. The system underperforms on counter-traditional configurations, such as female names featuring strong metallic radicals (钅) or male names featuring delicate literary markers.

5. Limitations and Future Work

While our spatial composition approach is highly predictive, several structural limitations remain:

- Flat Sequence Encoding: While the sequence contains IDCs, our 1D-CAE processes inputs as a flat sequence rather than explicitly assembling a recursive tree graph. Integrating a lightweight Tree-LSTM or Graph Convolutional network represents a valuable next step.
- Phonetic Blindness: The model lacks pronunciation vectors. Adding a parallel phonetic character branch (Pinyin phone-embeddings) would resolve phonetic-semantic clashes in given names.
- Vocabulary Coverage Limitations: Unseen characters absent from our 9,574 lookup database drop down to padding constants. Incorporating a universal dynamic Unihan decomposition dictionary would let the system parse any arbitrary logographic string instantly.

5.1 Proposed Empirical Ablation Framework

To definitively map and isolate the baseline mathematical value contributed by each localized morphological track in future extensions, we propose the execution of a comparative ablation matrix under a standardized parameter budget:

Model Variant	Radical Branch	Ordered Sequence	Pretraining	Frozen Encoder	Target Latent Dim
1. Character Token Baseline	—	—	No	—	64-D
2. Radical-Only Embedding	Yes	—	Yes	Yes	64-D
3. Component-Sequence Only	—	Yes	Yes	Yes	64-D
4. Radical + Unordered Bag	Yes	Unordered Bag	Yes	Yes	64-D
5. Proposed Architecture	Yes	Ordered IDS	Yes	Yes	64-D
6. Fine-Tuned System	Yes	Ordered IDS	Yes	No (Active)	64-D

Our hypothesis is that: (1) Ordered Structure > Unordered Bags because it preserves spatial assembly data, and (2) Pretrained Structural Embeddings > Random Embeddings due to embedded morphological rules.

6. Conclusion

This paper presents a lightweight and efficient transfer learning framework for logographic Chinese character strings in general, demonstrated effectively here on large-scale naming semantic tasks. By bypassing sequential LSTMs in favor of a parallelized 1D Convolutional Autoencoder and adopting a position-preserving feature concatenation layout, the model captures deep structural and morphological context without semantic signal dilution. The resulting pipeline handles approximately one million rows with a small RAM footprint, establishing a highly practical, flexible, and reusable benchmark for efficient CJK orthographic representation learning on consumer-grade hardware. The results provide an initial demonstration that a compact, frozen representation learned from Chinese sub-character structure can support downstream prediction. The proposed architecture provides a practical framework for evaluating how much transferable information can be retained in a compact Chinese character representation.

7. References

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